STRATEGY OVERVIEW · NQ, ES, RTY / MICROS

S30 — Cash-session IBS reversal

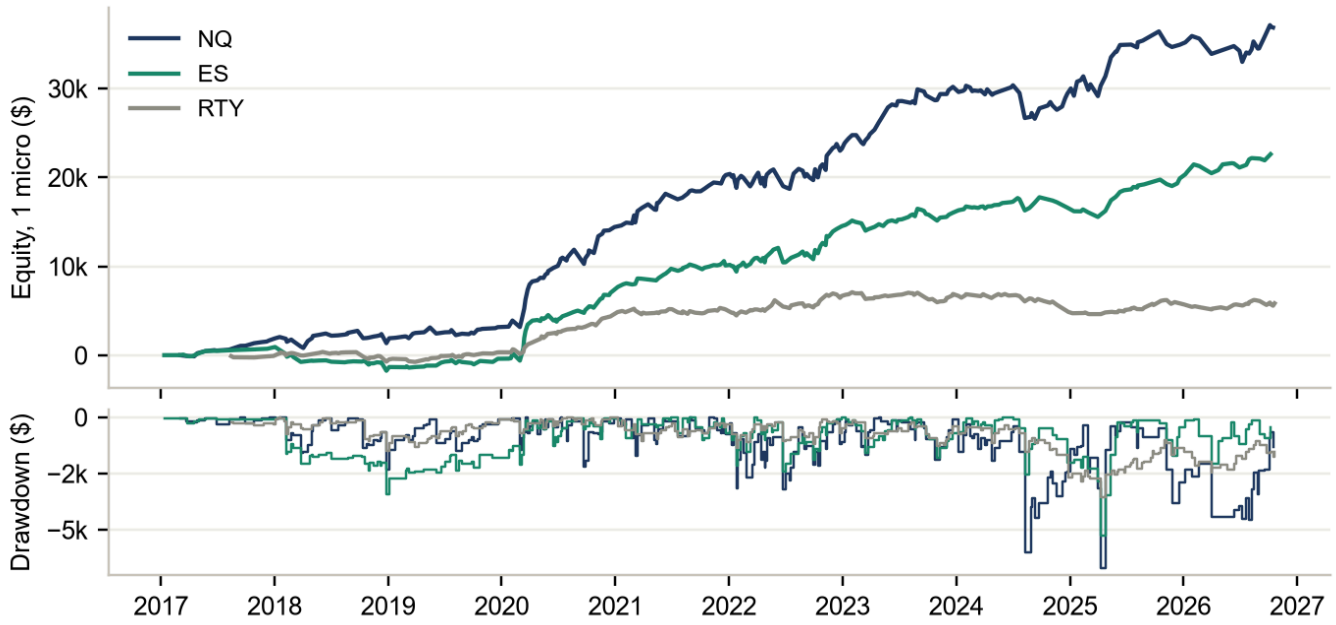
E-mini Nasdaq-100, S&P 500 and Russell 2000 · 15-minute bars · Chicago time · long only · one decision a day at the 15:00 CT cash close · holds overnight and over weekends · one contract per market

1 micro, whole backtest	Net	PF	Trades / yr	Win rate	Average	Max DD	Closed DD	Grade
NQ · MNQ	$36,868	1.86	28	70.3%	$135.05	$6,677	$3,764	B
ES · MES	$22,623	1.94	27	68.5%	$87.01	$5,238	$2,674	B
RTY · M2K	$5,852	1.30	29	58.7%	$21.59	$3,540	$2,463	not graded
All three	**$65,342**	**1.76**	**82**	**65.8%**	**$81.27**	—	**$5,785**	

MultiCharts backtests on 15-minute charts, first trade to the last closed trade: NQ 2017-03-02 → 2026-10-15 (273 trades), ES 2017-01-09 → 2026-10-07 (260), RTY 2017-08-08 → 2026-10-20 (271). $2.50 commission per contract and $5 slippage per trade. 1 micro = MNQ, MES or M2K (×10 = the full contract). Max DD counts closed equity plus each trade's worst open loss; "All three" shows closed-trade drawdown only, because its positions overlap. Positions still open at the end of the exports are not counted.

Roll correction: on the raw continuous charts, 62 trades were open across a quarterly contract roll and booked the price gap between contracts (NQ 17, ES 22, RTY 23; together $5,280 per micro, almost all of it profit). Every figure here has that gap removed.

Equity path and drawdown



The trade story

Internal bar strength (IBS) says where a session closed inside its own range: 0 at the low, 1 at the high. When an index future ends its cash session in the bottom fifth of the day's range, the day's selling ran right into the close. S30 buys just after the close and holds until a session closes in the top fifth of its range, usually a few days later (median hold 4 calendar days). The usual explanation is short-term overreaction: selling that has to be done by the end of the day (risk cuts, stops, hedges) pushes price too far, and whoever buys it is paid by the bounce. S30 does not measure who sells; it trades the published pattern on three index futures with one rule.

Rules

Session	The 08:30–15:00 CT cash session: 26 fifteen-minute bars on a full-session chart (bar stamps 08:45 … 15:00).
Signal	At the 15:00 CT bar: IBS = (close − session low) ÷ (session high − session low).
Entry	Flat and IBS < 0.20: buy one contract at market on the next bar (15:00 CT; the futures keep trading).
Exit	Long and IBS > 0.80: sell at market on the next bar.
Nothing else	No stop, target, filter or time exit. Positions are held through nights, weekends, holidays and contract rolls. A session with missing bars or no range makes no decision. The same file runs on @NQ, @ES and @RTY, one chart each.

By year, 1 micro

Year	NQ trades	NQ net	ES trades	ES net	RTY trades	RTY net	All three
2017	16	$1,699	16	$855	6	−$64	**$2,490**
2018	27	−$324	24	−$2,582	28	−$633	**−$3,539**
2019	24	$1,804	21	$1,346	29	$1,051	**$4,202**
2020	36	$11,254	36	$7,407	31	$4,027	**$22,688**
2021	29	$5,850	28	$3,084	31	$786	**$9,720**
2022	41	$2,953	39	$4,424	38	$1,520	**$8,898**
2023	31	$6,960	32	$1,513	31	$216	**$8,689**
2024	29	−$753	26	$562	25	−$2,016	**−$2,207**
2025	23	$5,486	22	$3,282	30	$1,104	**$9,872**
2026	17	$1,940	16	$2,731	22	−$141	**$4,530**

By exit year; 2017 from each market's first trade, 2026 to the last closed trade. Positive years: NQ 8 of 10, ES 9 of 10, RTY 6 of 10.

Worst drawdowns (including open-trade loss), 1 micro

Market	Depth	Peak	Low	Recovered	Days
NQ	$6,677	2025-02-12	2025-04-09	2025-04-24	71
NQ	$6,014	2024-07-01	2024-08-08	2025-01-23	206
ES	$5,238	2024-09-24	2025-04-09	2025-05-27	245
ES	$3,430	2018-01-02	2018-12-26	2020-03-13	801
RTY	$3,540	2023-01-31	2025-04-09	not yet	1358+
RTY	$1,480	2018-06-04	2018-12-26	2020-02-05	611

Now from the peak: NQ −$265, ES $0, RTY −$1,252. Largest single loss: NQ −$2,814, ES −$1,630, RTY −$804; worst open loss inside one trade: NQ −$5,148.

Why it should work

- **Published effect:** Pagonidis, "The IBS Effect: Mean Reversion in Equity ETFs" (NAAIM, 2014): equity ETFs averaged +0.35% the next day after an IBS below 0.20 (before costs, trading at the close). The exit above 0.80 is the practitioner version of the rule (Murtazin, 2025).
- **Run once on later data:** the thresholds are the source's, not fitted. After a 2017–2022 run the rule was frozen and run once on 2023-01-03 → 2026-08-10 in research: ES PF 2.06, NQ 1.69, RTY 1.02 (ES's 2023+ result had been seen once before, through the ES-only version). In MultiCharts on 2023+ the ES and NQ legs repeated it (PF 2.04 and 1.67).
- **Two-sided:** the mirror trade (short above 0.80 until below 0.20, measured on 2017–2022, not traded) also made money: ES PF 1.28, NQ 1.35, RTY 1.36. The reversal is not just a long bias in a rising market.
- **2022, a falling year:** all three made money while holding lost (per micro: NQ $2,953 vs −$10,900, ES $4,424 vs −$4,615, RTY $1,520 vs −$2,439).

Risks and caveats

- **In rising years it earns about what holding would:** in the 2023–26 research run each leg made roughly what holding the index for the same time in the market made (NQ $133,940 vs $131,100, ES $90,442 vs $69,600 per full contract). The edge there shows as a higher PF and a smaller drawdown, not as extra return.
- **2018 did not repeat 2022:** all three lost; ES lost more than holding (−$2,582 vs −$928 per MES), NQ about matched it, RTY lost less.
- **RTY is the weak leg:** PF 1.69 on 2017–2022 but 1.02 on 2023–26 in research ($1,880 per full contract against $26,900 for holding).
- **No stop by design:** single losses reach −$2,814 per MNQ (−$28,135 per NQ).
- **Overnight, weekend and roll risk:** positions are held for days; not for accounts that must be flat each day. A position open at a roll has to be rolled to the next contract live.
- **Execution:** the published effect is measured at the close; a later study found an ETF version fell apart when traded open-to-open (Pandey and Joshi, 2023 preprint). S30 trades at 15:00 CT, right after the cash close.
- **Roll gaps** were measured on dxFeed data at each splice. Two are estimates: June 2022 (the jump also held a weekend sell-off; the average of the neighbouring rolls is used) and ES September 2026 (no minute data; the June 2026 roll's size in percent is used).

Robustness grade

Split at 2023-01-01, 1 micro	NQ B	ES B	RTY
Trades in / out of sample	173 / 101	164 / 96	not graded
Expected value per trade, in → out	$136.41 → $157.30	$87.97 → $102.30	—
Kept out of sample · P(edge ≤ 0)	115% · 0.0155	116% · 0.0023	—
Out-of-sample net / max DD (closed)	$15,888 / $3,764	$9,821 / $1,863	—
Return ÷ drawdown, out of sample	4.22	5.27	—
Monte-Carlo max DD, median / 75th / 95th / 99th	$4,482 / $5,658 / $7,966 / $9,979	$2,421 / $3,038 / $4,270 / $5,459	—

Graded 2026-10-17 on earlier exports with the same trades (NQ's then-open last trade included; the ES grade on the identical ES-only version of this rule), before the roll correction. Expected value from a 10,000-draw bootstrap; drawdowns from a 20,000-path Monte Carlo of the trade sequence (closed equity). Grade scale S–D combines edge kept out of sample, confidence it is above zero, and return ÷ drawdown.

Research and earlier MultiCharts runs, 1 micro

	Research · 2017-07-10 → 2022-12-30	Research · 2023-01-03 → 2026-08-10 (once)	MultiCharts · 2023 →
NQ	166 trades · PF 2.07 · $23,058	94 · PF 1.69 · $13,394	96 · PF 1.67 · $13,514 (to 2026-09-01)
ES	153 · PF 1.98 · $13,925.50	93 · PF 2.06 · $9,044.20	94 · PF 2.04 · $8,979 (to 2026-09-11)
RTY	163 · PF 1.69 · $6,747	101 · PF 1.02 · $188	—

Research = computer simulation from 1-minute bars at $15 per round trip; the rule was frozen before the 2023+ run. None of these earlier runs remove roll gaps. "—" = not run.

Tried and rejected — do not repeat

When	Tried	Result	Verdict
2026-09-16	Also require the close to be below the session VWAP	Identical trades on 2017–2022: a close in the bottom fifth of the range is always below VWAP	Adds nothing
2026-09-16	Short side (short above 0.80, cover below 0.20), measured on 2017–2022	ES PF 1.28, NQ 1.35, RTY 1.36	Not traded; the source rule is long only
2026-09-16	Fourth market: Nikkei 225 (NKD)	Research: nothing on 2017–2022 (PF 1.00), then +$102,071 per contract on 2023–26; the MultiCharts run did not pay	Dropped
2026-10-23	Backtest on the raw continuous chart, positions carried through rolls	62 trades booked the gap between contracts as profit or loss	Corrected here

Kill test — what would prove it wrong

Watch	From the data, 1 micro	Warning sign
Drawdown	Worst so far NQ $6,677, ES $5,238, RTY $3,540; Monte-Carlo 75th NQ $5,658, ES $3,038	Beyond the 75th percentile.
Win rate vs the win rate it needs	NQ 70.3% vs 56.1% · ES 68.5% vs 52.8% · RTY 58.7% vs 52.4%	Win rate sliding toward the needed level.
Trade count	≈ 28 a year per market	Materially different: data or session setup changed.
Falling markets	2022: all three made money while holding lost; 2018: only ES lost more than holding	A falling year where it loses more than holding on two or more markets.

Median hold: NQ 4, ES 4, RTY 4 calendar days. Largest win: NQ $2,142, ES $1,195, RTY $707. Positive months: NQ 79/116, ES 77/118, RTY 61/111. Source: MultiCharts backtests; rules from the strategy code. Backtests, not account results. Prepared 2026-10-23.